

CricPulse UI/UX Analysis

HCI Project

Team 12

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1 Introduction

1.1 Project Overview

CricPulse is a fully functional online cricket portal and live scoring application. The main goal of this project is to build an interactive website that puts into practice the Human-Computer Interaction (HCI) concepts we learned in class. Instead of just designing static screens, I wanted to build a working system that focuses on giving users control, making the site easy to learn, avoiding mental overload, and ensuring it is accessible to everyone, from casual fans to hardcore statisticians. To achieve this, the original prototype was massively expanded to a comprehensive 20-page interactive environment.

1.2 Objective and Scope

To make a sports app that people actually want to use, it has to be more than just visually appealing. In CricPulse, I tried to bridge the gap between classroom theories—like Gestalt psychology, Fitts’s Law, Shneiderman’s Eight Golden Rules, and Nielsen’s Heuristics—and actual web development. I also used the Vital Few (80-20 Rule), meaning I focused most of my effort on the core 20% of features that make up 80% of the actual user experience. By following standard Web Usability Guidelines, the design matches what users already expect. Thanks to the Law of Learning, this means users won’t need to spend any time figuring out how to use the app because it already feels familiar to them.

2 Live Dashboard and Real-Time Data Visualization

The core experience of any sports platform revolves around immediate access to ongoing matches. It is critical that users can instantly consume live data and understand system status without visual clutter or latency anxiety.

2.1 Information Architecture via Inverted Pyramid

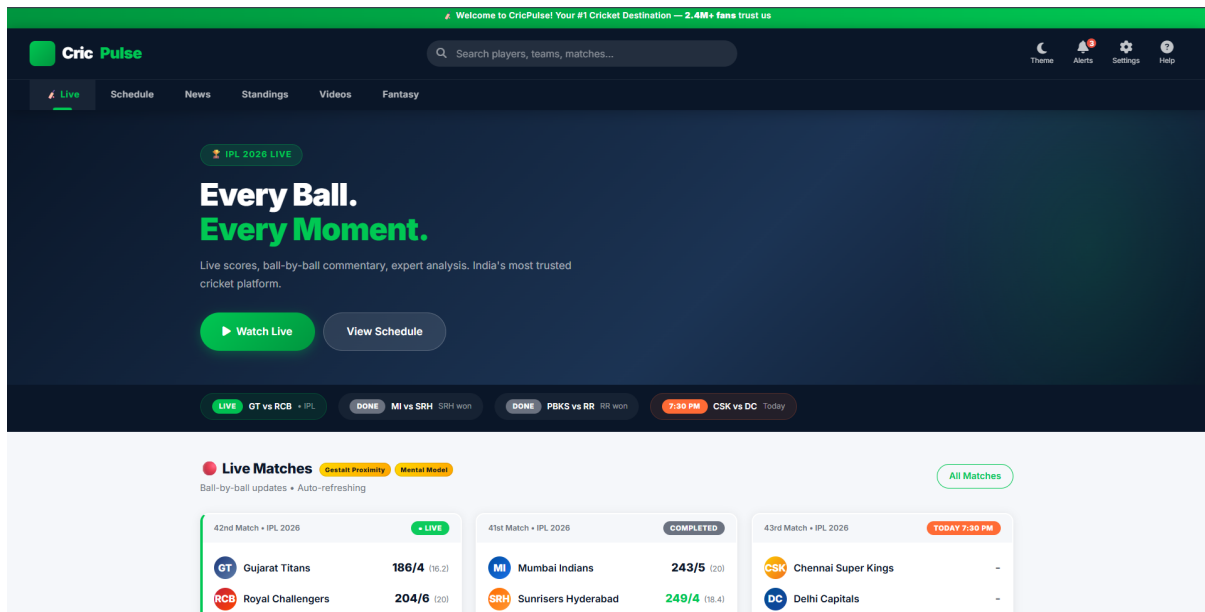


Figure 1: Home Page using Inverted Pyramid layout

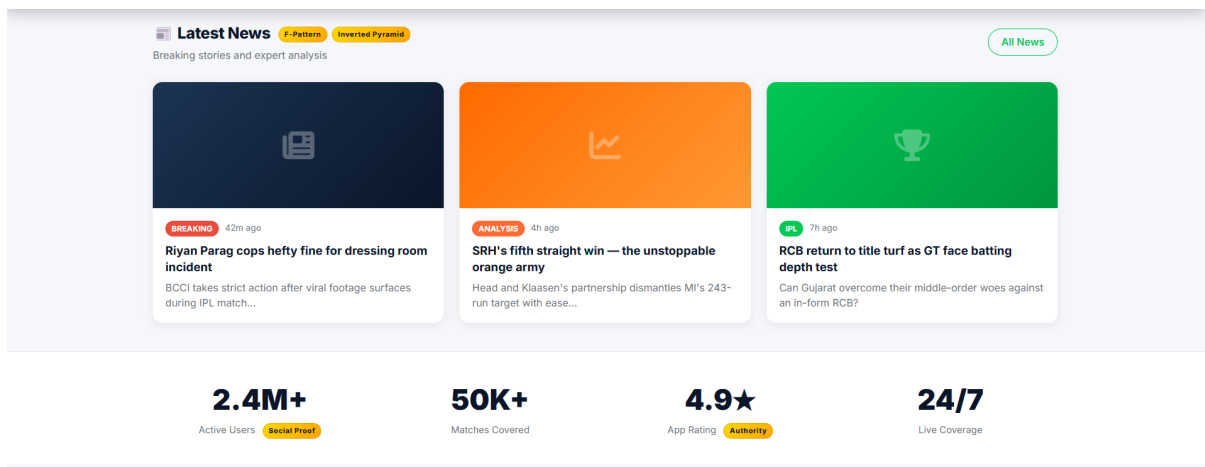


Figure 2: Home Page

Relying on a traditional, flat layout for sports news is often insufficient during live events. CricPulse fixes this by structuring the Home dashboard using the Inverted Pyramid model. The page layout follows basic Visual Design principles of Balance, Scale, and Dominance. A large, high-fidelity Live Match Ticker acts as the main focal point at the absolute top of the screen. This leverages the Serial Position Effect (Primacy Effect); users immediately see active scores upon loading. Furthermore, the design incorporates Social Proof by highlighting "2.4M+ Active Users," building immediate trust and ensuring the viewing experience is much smoother. The use of whitespace prevents cognitive overload, ensuring the high-density numerical data is parsed easily.

2.2 Visibility of System Status and Gestalt Closure

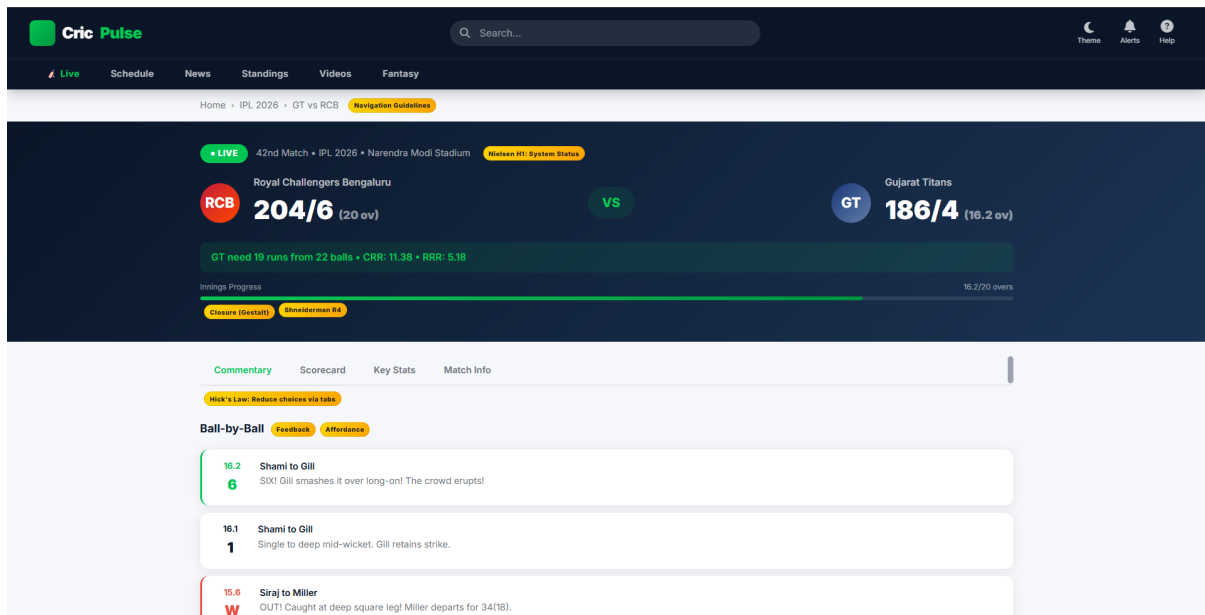


Figure 3: Match Detail page providing Visibility of System Status

During a live match, users must trust that the data is current. The Match Detail page gives the user clear Visibility of System Status and shows Robustness when handling live updates. A pulsating "LIVE" badge and dynamic timestamps are persistently displayed, satisfying Nielsen's first heuristic. Furthermore, the live innings progress bar utilizes the Gestalt Principle of Closure. By displaying a partially filled bar, the user's brain naturally closes the remaining empty space, allowing them to instantly conceptualize the remaining match time. To prevent cognitive overload from massive data streams, the page uses Hick's Law by hiding secondary data behind horizontal tabs.

3 Content Discovery and Temporal Navigation

Users frequently navigate the platform to find post-match editorials or upcoming fixtures. This requires an architecture that supports natural eye scanning and chronological comprehension.

3.1 F-Pattern Layout and Gestalt Similarity

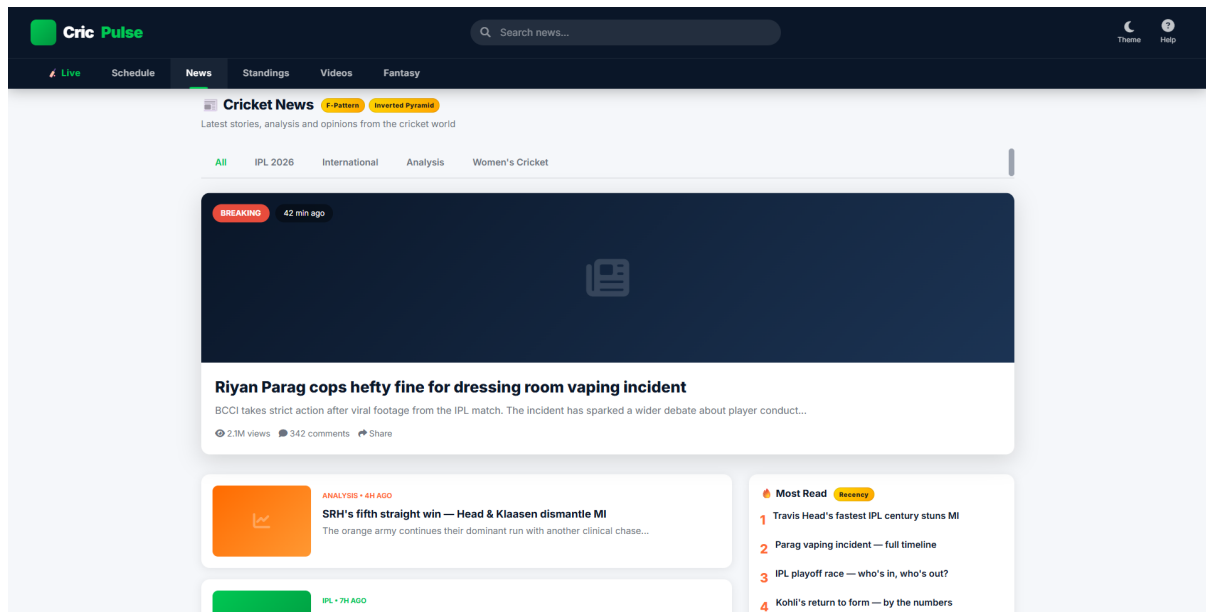


Figure 4: News Feed mapped to F-Pattern scanning

Reading through huge blocks of text or endless news feeds can be exhausting. To keep the screen clean and scannable, the News Feed is structurally engineered to support the F-Pattern reading behavior. A massive "Featured Story" anchors the top left, utilizing Visual Dominance to catch the eye. As the user scans down, secondary articles are aligned in a strict grid. The UI also relies on the Gestalt Law of Similarity; articles belonging to the same tournament share identically colored badges, allowing users to rapidly filter content visually without interacting with any explicit UI controls. The sidebar leverages the Recency Effect by surfacing "Trending Now" articles, catering to users who want immediate, high-velocity content updates.

3.2 Progressive Disclosure and Gestalt Proximity

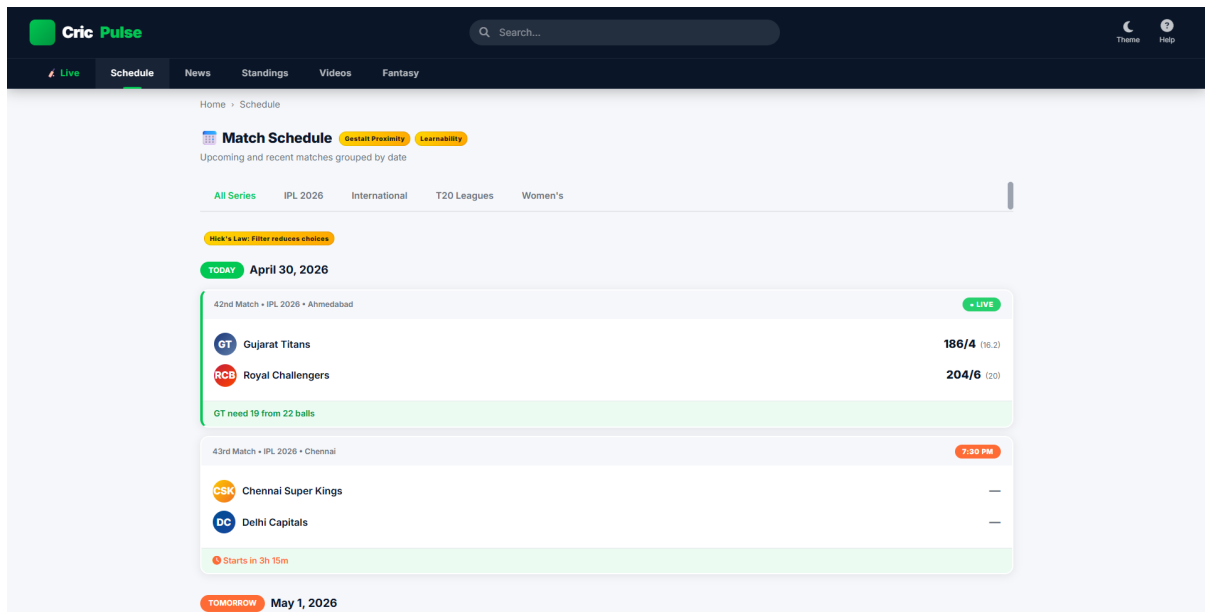


Figure 5: Schedule organized via Gestalt Proximity

Presenting a continuous list of dozens of upcoming matches is highly intimidating. To solve this, the schedule uses Gestalt Proximity. Match cards are grouped tightly together under prominent Date headers. Because the individual cards are physically closer to their specific date header, the user effortlessly perceives them as discrete, daily blocks. Furthermore, the page applies Hick's Law via Progressive Disclosure; category filters at the top provide a low-friction mechanism to instantly eliminate irrelevant series, significantly reducing the volume of data presented on the screen. The consistency of these cards across the platform ensures the Law of Learning is respected.

4 Statistical Complexity and Data Representation

Cricket generates an overwhelming amount of statistical data. Representing this data accurately while maintaining readability is a primary HCI challenge.

4.1 Visual Balance and Recognition over Recall

Home > IPL 2026 > Points Table

Points Table Visual Balance Gestalt Similarity

IPL 2026 — Updated after Match 41

Qualified / Top 4 Eliminated Win Loss Nielsen HE: Recognition

#	TEAM	M	W	L	NR	NRR	PTS	FORM
1	Sunrisers Hyderabad	9	7	2	0	+1.284	14	W W W W W
2	Royal Challengers	9	6	3	0	+0.856	12	W L W W W
3	Chennai Super Kings	8	6	2	0	+0.643	12	W W L W W
4	Rajasthan Royals	9	5	4	0	+0.321	10	W L W L W
5	Kolkata Knight Riders	8	5	3	0	+0.154	10	L W W W L
6	Gujarat Titans	8	4	4	0	-0.128	8	L L L W L
7	Delhi Capitals	8	4	4	0	-0.342	8	L L W W L
8	Lucknow Super Giants	8	3	5	0	-0.456	6	W L L L W
9	Punjab Kings	9	2	7	0	-0.923	4	L L L W L
10	Mumbai Indians	9	2	7	0	-1.105	4	L L L L L

Figure 6: Standings table utilizing Visual Balance

Data tables can easily cause cognitive fatigue. The Points Table uses strict Visual Balance rules: alphabetic team names are left-aligned, while numerical statistics are center-aligned, creating a predictable grid. To support Nielsen's heuristic of Recognition over Recall, the page provides a persistent visual Legend explaining the abstract color coding (e.g., green 'W' dots for wins). The design also uses the Gestalt Law of Similarity by inserting a dashed line and highlighting the top 4 teams, grouping them as a "successful" subset and communicating their structural advantage without requiring the user to read rank numbers.

4.2 Conservation of Complexity and Figure-Ground Relationship

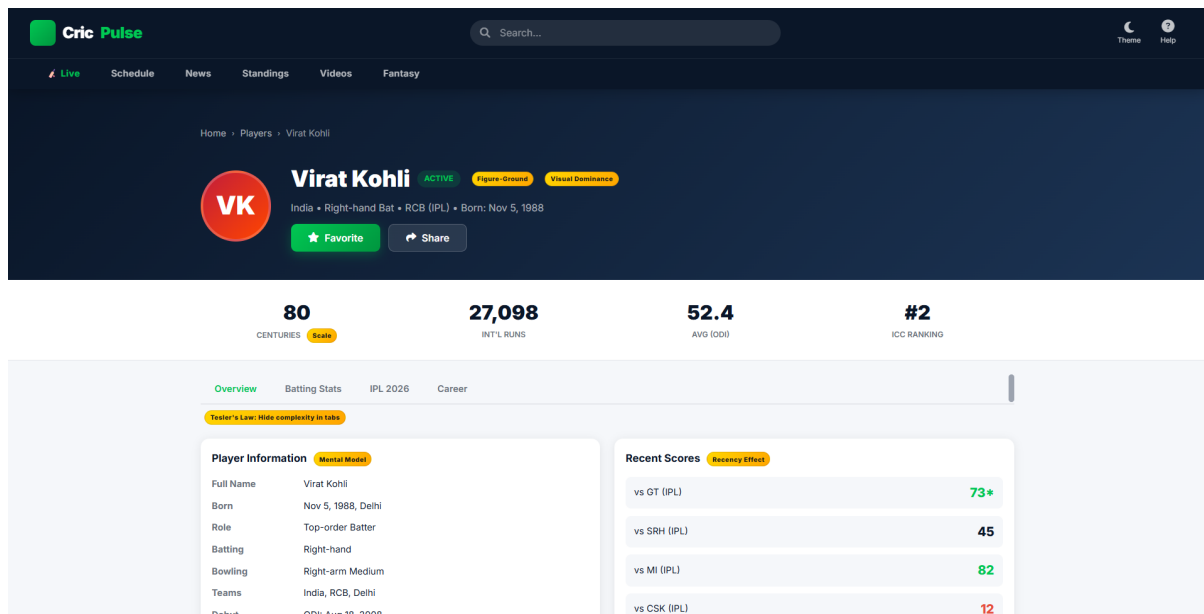


Figure 7: Player Profile applying Tesler's Law

A professional athlete's career generates an irreducible amount of data. According to Tesler's Law (Conservation of Complexity), this complexity cannot be eliminated, only shifted. The UI shifts the burden to the architecture by compartmentalizing data into selectable tabs, hiding the complexity until it is explicitly requested. The hero section utilizes a strong Figure-Ground Relationship, placing the brightly colored player portrait against a deep gradient background to ensure critical identifying information stands out. Historical milestones are presented along a vertical timeline, employing the Gestalt Law of Continuity.

5 Search Mechanics and Multimedia Interaction

Facilitating rapid access to specific entities and handling multimedia interactions requires robust, error-tolerant design patterns.

5.1 Fitts's Law and Error Recovery

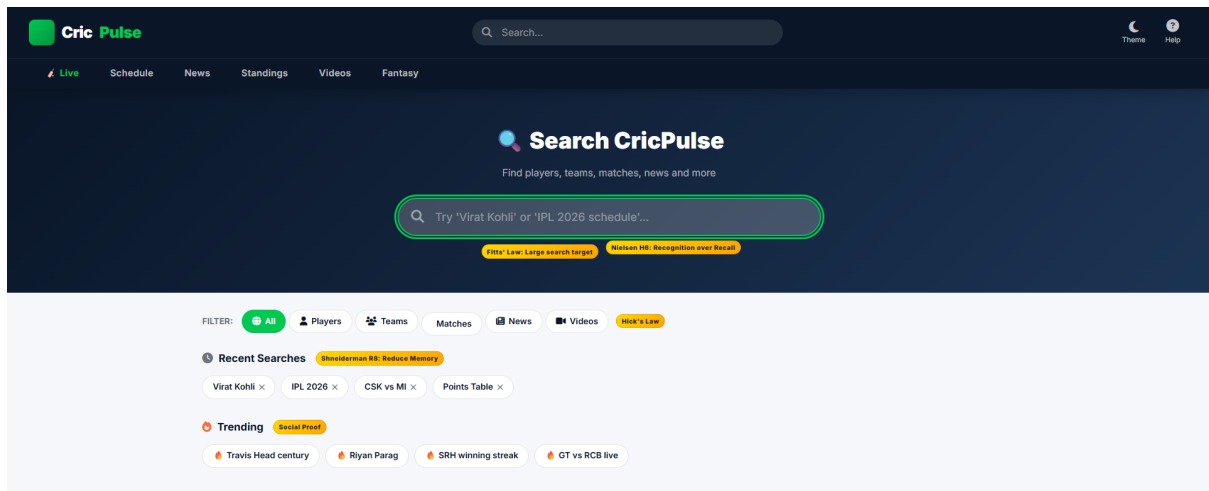


Figure 8: Search interface optimizing Fitts's Law

Search is a primary utility that requires immediate access. The main search input field is designed to be massive, occupying a significant percentage of the viewport. This dramatically increases the target area, optimizing Fitts's Law by reducing the motor precision required to initiate a query. To reduce short-term memory load (Shneiderman's 8th Rule), "Recent Searches" are automatically displayed. If a query fails, the UI gracefully handles it by presenting a friendly "No results found" graphic with plain-language advice and primary action buttons, adhering to Nielsen's Error Recovery heuristic.

5.2 Direct Manipulation and Visual Rhythm

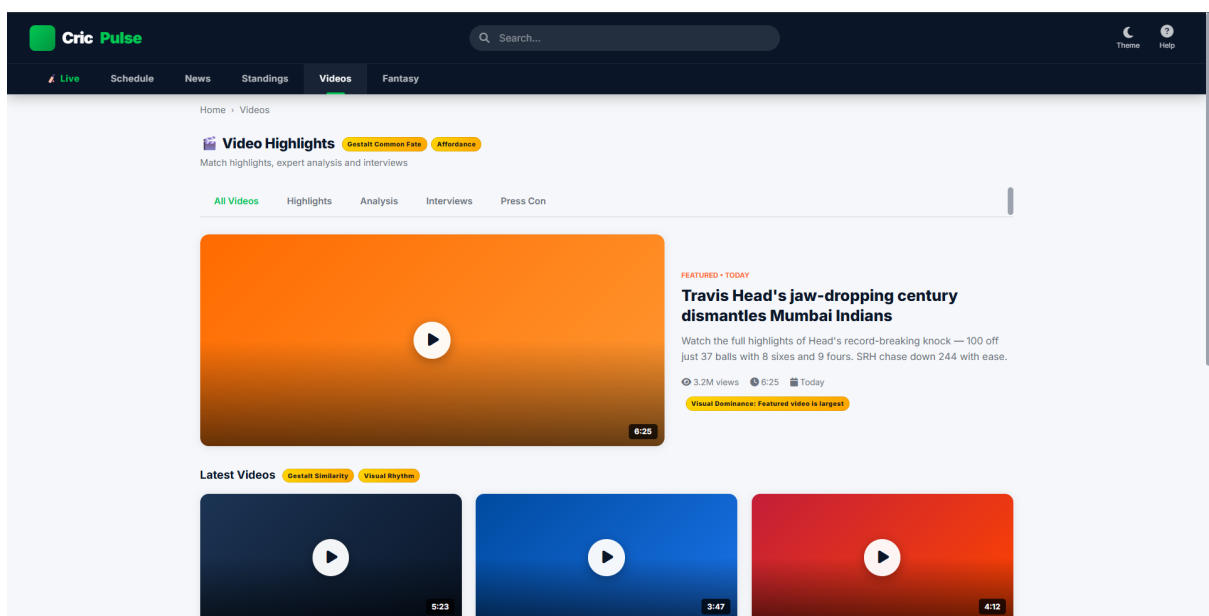


Figure 9: Video Highlights gallery demonstrating Direct Manipulation

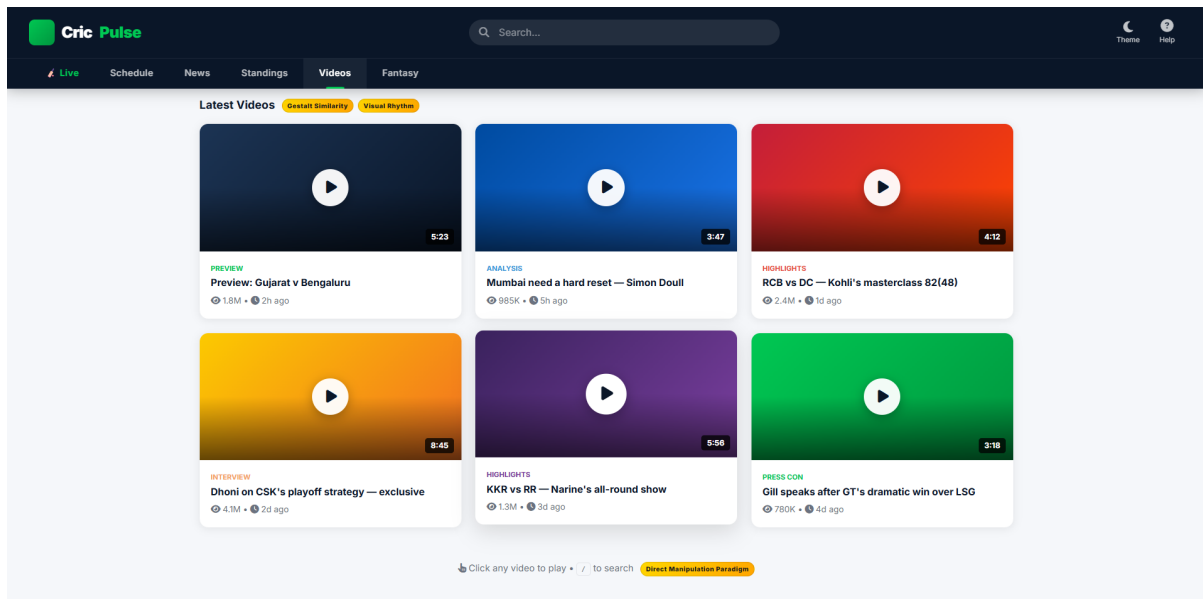


Figure 10: Video Highlights page

Video thumbnails require strong visual cues to indicate their specific interactivity. Every thumbnail is overlaid with a prominent "Play" icon that scales up physically on hover. This strong affordance maps to the Direct Manipulation paradigm, mimicking interaction with a physical media player. The secondary gallery establishes a predictable Visual Rhythm across a strict grid. When filtering categories, irrelevant cards faded out while the remaining cards animate into new positions together, applying the Gestalt Law of Common Fate so the perceptual system instantly groups them as a unified subset.

6 Gamification and Persuasive Design Patterns

To sustainably drive user retention beyond simply checking scores, the platform introduces gamification and ethical persuasion mechanics.

6.1 Form-Fill Paradigms and Cognitive Closure

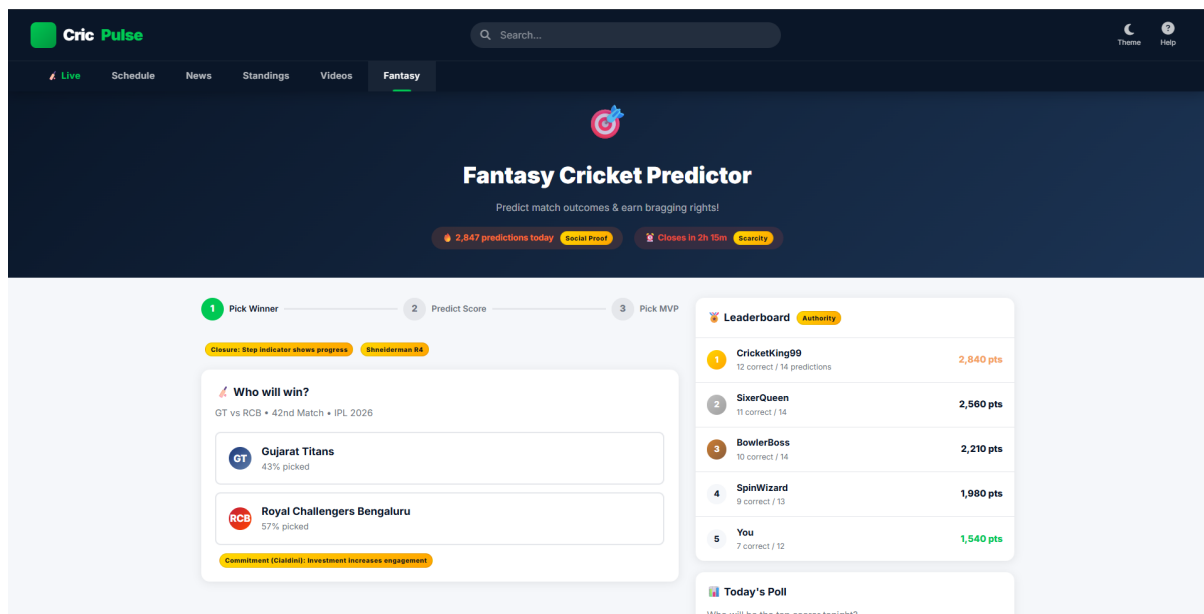


Figure 11: Fantasy Predictor guiding user via Cognitive Closure

The gamified Fantasy Predictor is architected as a sequential, multi-step form. A prominent 1-2-3 step indicator is anchored at the top. This directly addresses Shneiderman's rule to design dialogs to yield closure, visually communicating exact progress. By breaking the prediction into steps, the UI leverages the principle of Commitment. Step 1 is a low-friction choice; once made, the user is psychologically committed and more likely to complete higher-friction tasks. A countdown timer creates artificial urgency, implementing the psychological principle of Scarcity.

6.2 Social Proof and Extrinsic Motivation

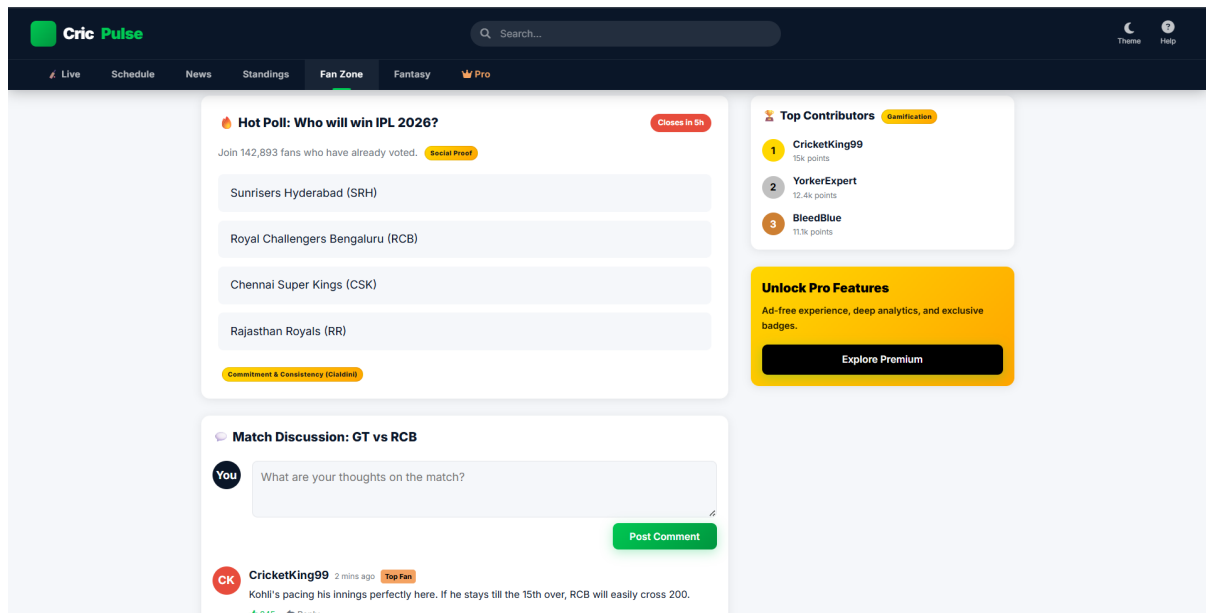


Figure 12: Fan Zone driving Extrinsic Motivation

The Fan Zone introduces systemic gamification to foster community engagement. A leaderboard assigns points to interactions, displaying a public ranking system that taps into Extrinsic Motivation. The live polling interface explicitly highlights the massive scale of participation (e.g., "142,893 fans voted"). By doing so, the system leverages Social Proof; seeing peers participate significantly increases the likelihood of user conformity. The comment thread architecture relies on the Law of Reciprocity, where receiving "likes" creates a socially validated positive feedback loop that compels the user to return.

7 Universal Design and Support Architectures

Good design needs to take different physical and cognitive abilities into account right from the start, providing explicit control mechanisms.

7.1 Flexibility in Use and Locus of Control

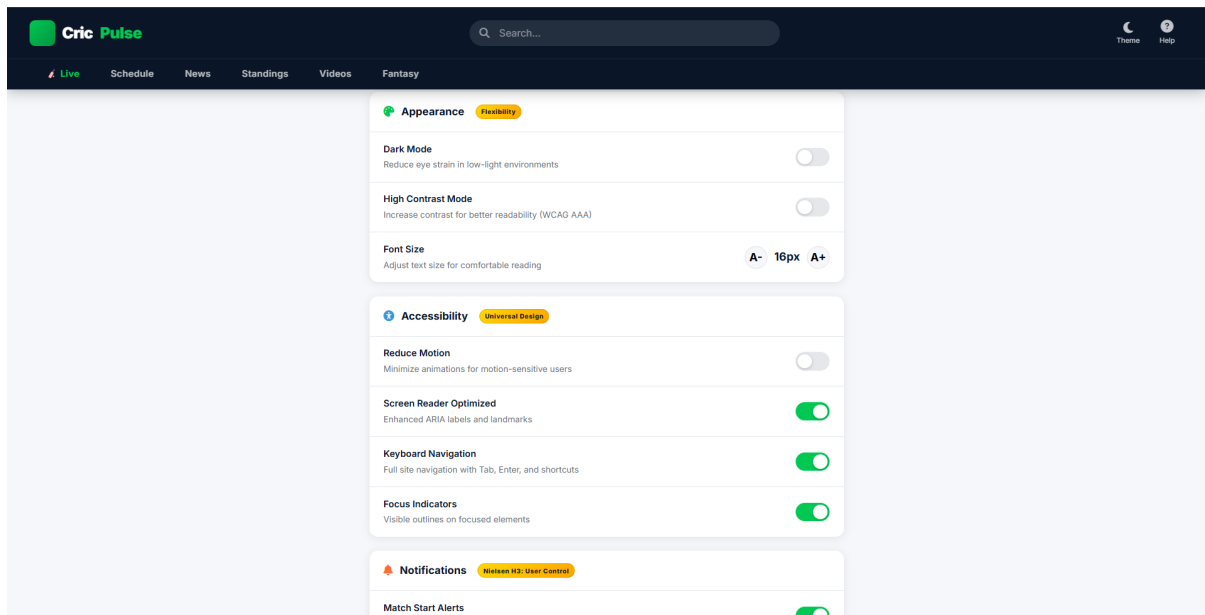


Figure 13: Settings page ensuring an Internal Locus of Control

Users usually dislike apps that don't let them change their preferences. To create a strong Internal Locus of Control, the Settings page provides granular UI toggles for notification categories and explicit buttons to clear history. This provides a lot of Flexibility, allowing the app to adapt. The platform strictly adheres to the Seven Principles of Universal Design. Hardware-accelerated toggles for Dark Mode, High Contrast Mode, and dynamic Font Size scaling ensure maximum usability for all demographics, satisfying the requirement for Equitable Use.

7.2 Architectural Implementation of User Support Systems

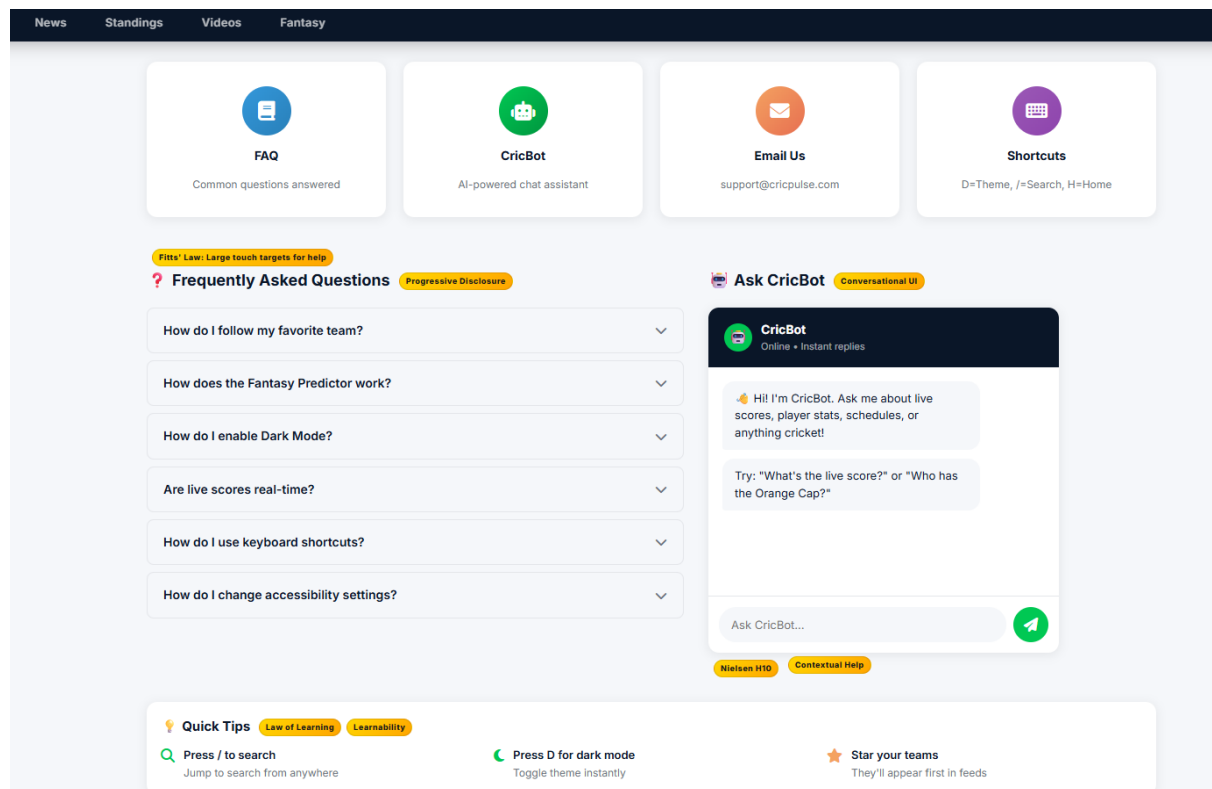


Figure 14: Help page built with explicit User Support Systems Principles

While the system strives for intuitive use, it is critical to implement robust User Support Systems. Nielsen's 10th heuristic requires accessible documentation for edge cases. The Help page centralizes FAQs, direct email support, and keyboard shortcuts. Instead of a massive wall of text that is hard to read, the FAQs adhere to User Support Systems Principles by utilizing Progressive Disclosure via an accordion pattern; answers are hidden until explicitly clicked, preventing information overload. Furthermore, the "CricBot" feature implements a modern Conversational User Interface, allowing natural language queries. This masks complex database retrieval behind a familiar chat interaction, drastically lowering the barrier to entry for finding support.

8 Notifications and Monetization Strategies

Handling system alerts and financial transactions requires a UI built entirely on trust, immediate feedback, and transparent framing.

8.1 Action Reversal and Informative Feedback

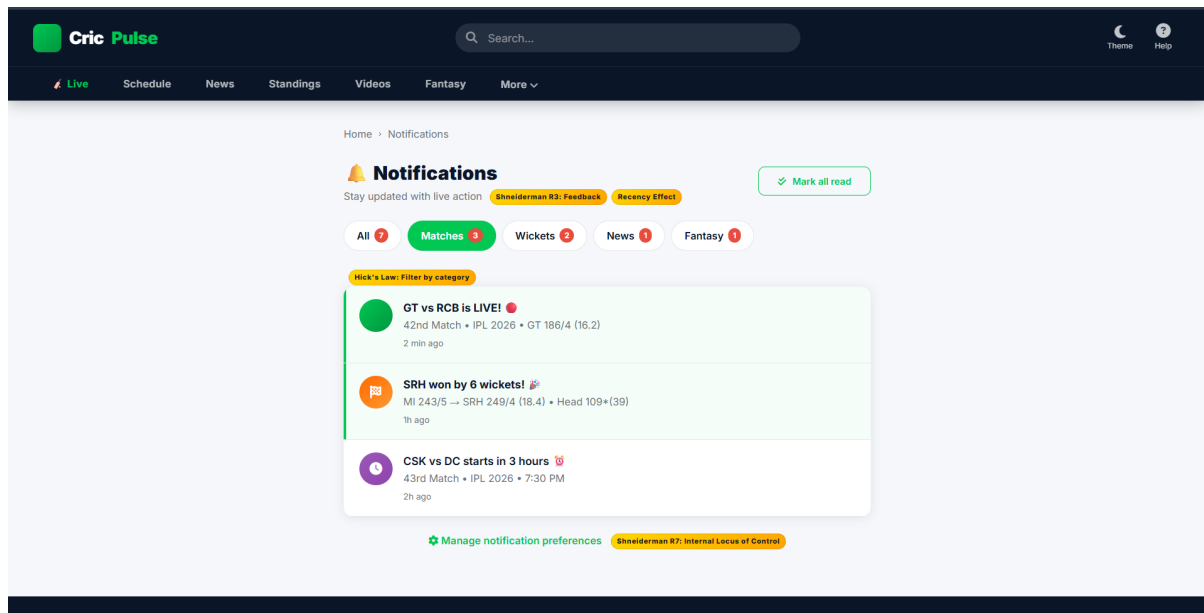


Figure 15: Notifications Center providing Informative Feedback

The Notifications Center acts as a chronological ledger. It provides Informative Feedback (Shneiderman R3) by applying a subtle tinted background to unread items, instantly communicating what requires attention. When a user dismisses an alert, the item collapses and triggers a temporary toast offering an explicit "Undo" button. Adhering to Shneiderman R6, knowing that a potentially destructive action can be instantly reversed relieves user anxiety. The list is strictly ordered to align with the Recency Effect, forcing the newest data to the top of the visual stack.

8.2 Anchoring Effect and Visual Dominance

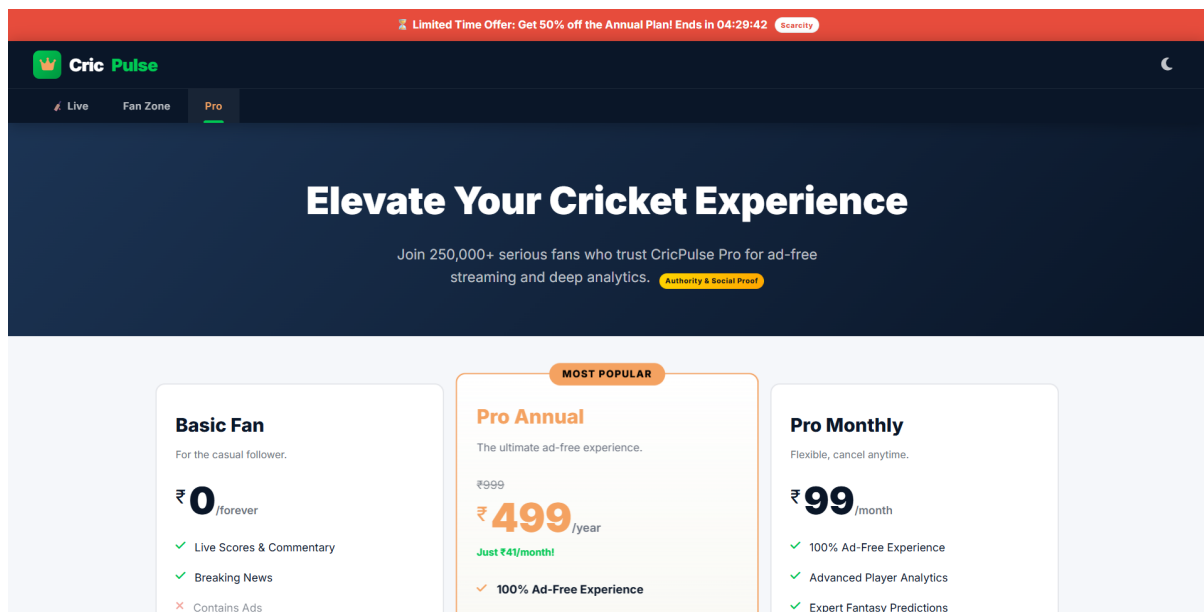


Figure 16: Premium Subscription utilizing the Anchoring Effect

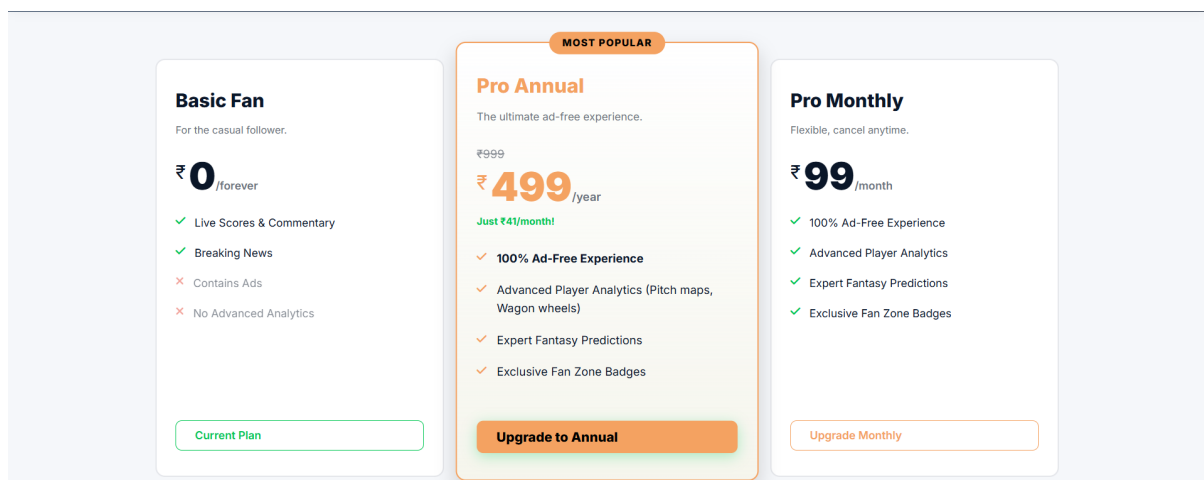


Figure 17: Premium Subscription page

The Premium page handles monetization by relying on persuasive design patterns. The pricing matrix places a free tier adjacent to paid tiers, and the Annual plan displays a visible "strikethrough" price. This utilizes the Anchoring Effect, making the actual price subjectively appear as a significantly better deal. The UI attempts to steer the user toward the Annual subscription by physically enlarging the card and utilizing a gold border. Through heavy Visual Dominance, the interface frames this specific choice as the default and most logical option. Finally, the polished aesthetic inherently validates the platform's authority to process payments.

9 Identity Management and Historical Archives

Expanding the platform requires robust identity management and methods for navigating massive historical datasets without confusing the user.

9.1 Authentication and Cognitive Load Reduction

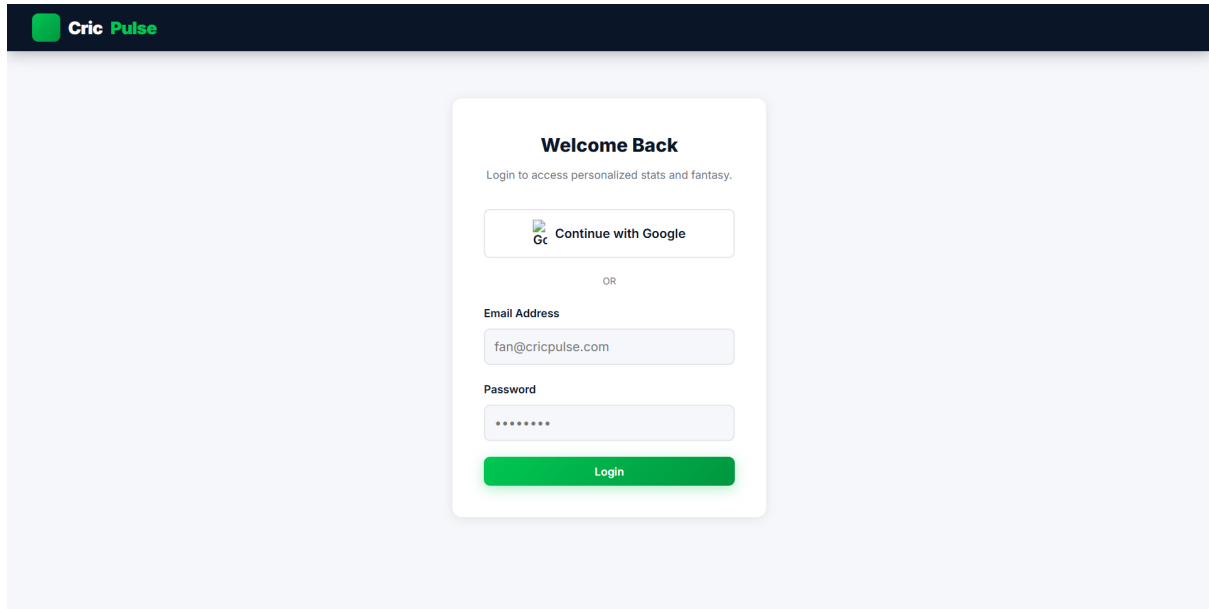


Figure 18: Authentication page minimizing Cognitive Load

The Login and Signup pages are critical bottlenecks in user acquisition. To minimize Cognitive Load, the interface heavily emphasizes "Single Sign-On" (SSO) options like Google or Apple login. By prioritizing these one-click authentication methods over manual form entry, the system respects the user's time and prevents drop-off. The form fields themselves utilize inline validation (Error Prevention), ensuring users do not submit incorrectly formatted emails, which saves them the frustration of waiting for a server response to realize a mistake was made.

9.2 Information Retrieval and Hick's Law Filters

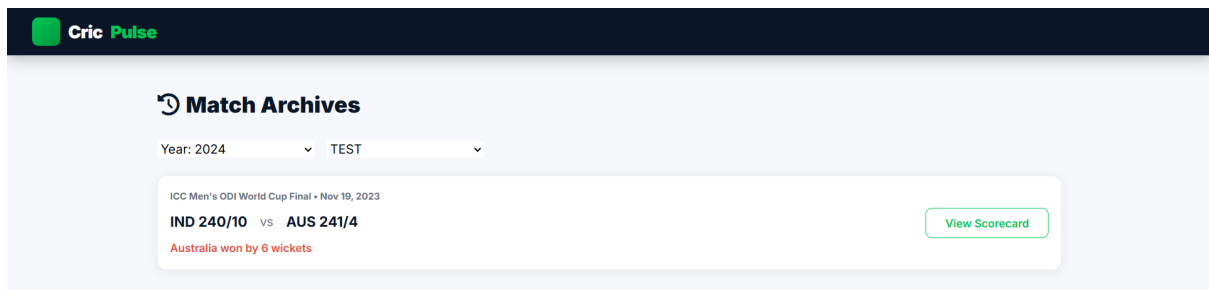


Figure 19: Archives page utilizing Hick's Law Filters

The Match Archives page provides access to decades of historical cricket data. Presenting this in a flat list is impossible. Therefore, the interface relies heavily on Hick's Law by implementing robust, cascading dropdown filters (Year, Format, Tournament). Before displaying any data, the user is required to narrow their search parameters. This dramatically reduces the number of choices and visual stimuli presented on the screen, converting a massive database query into a highly manageable, targeted list of relevant past matches.

10 E-Commerce and Scarcity Metrics

Integrating physical merchandise and ticket sales requires specific e-commerce HCI paradigms to ensure seamless conversions.

10.1 Ticket Booking and Artificial Scarcity

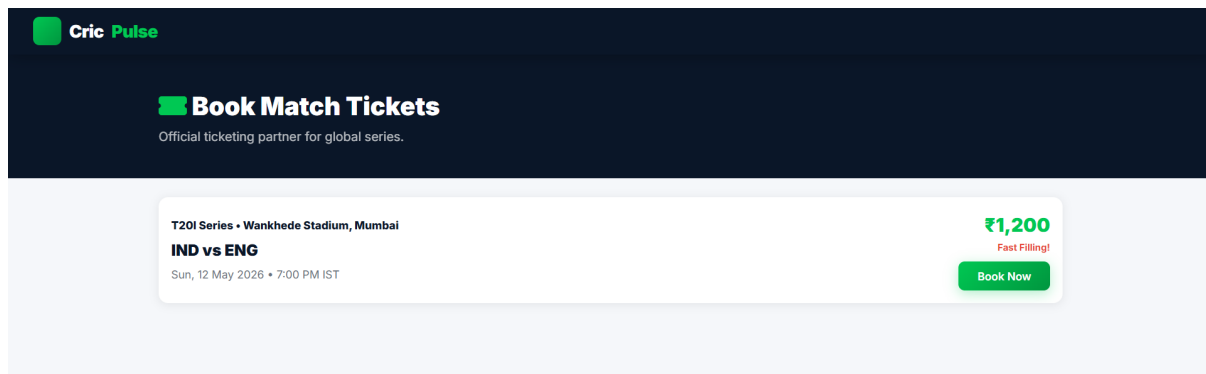


Figure 20: Ticket Booking page using Artificial Scarcity

When selling live event tickets, driving immediate action is paramount. The Ticket Booking interface utilizes the psychological principle of Scarcity. Alongside the price and match details, the UI dynamically injects microcopy such as "Fast Filling!" or "Only 4 seats left in this block" rendered in high-alert red colors. This creates artificial urgency, forcing the user to bypass prolonged decision-making paralysis and execute the purchase immediately due to the Fear of Missing Out (FOMO).

10.2 Store Interface and Dual-Coding Theory

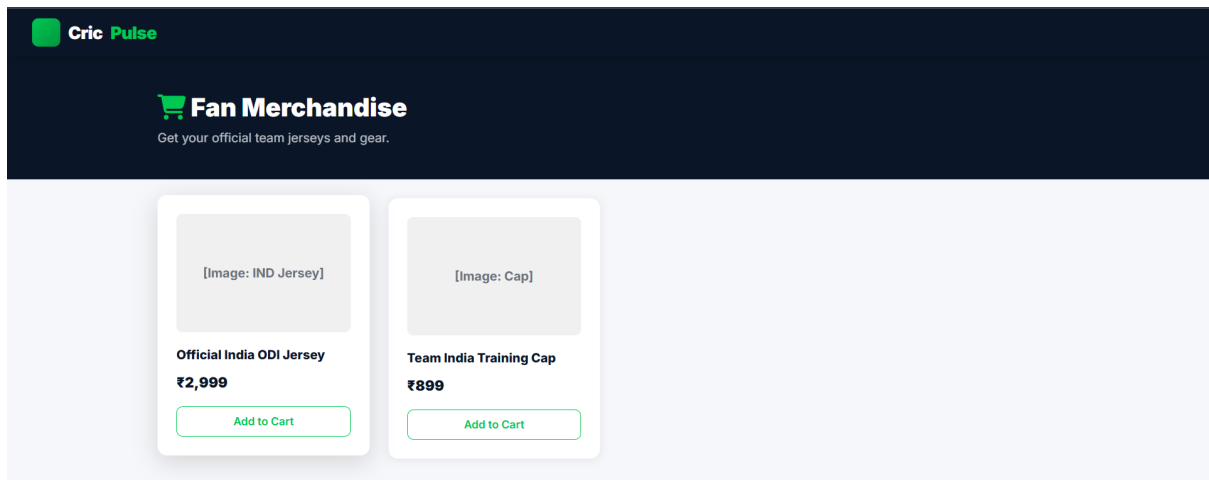


Figure 21: Merchandise Store applying Dual-Coding Theory

The official merchandise store relies entirely on Dual-Coding Theory. Text descriptions of physical items (like jerseys or caps) are insufficient for conversion. The interface places massive, high-fidelity images directly above the product titles and bold pricing. By providing both visual and verbal representations of the information simultaneously, the user processes the product data much faster. The stark, high-contrast "Add to Cart" buttons serve as massive call-to-action targets, ensuring Fitts's Law is respected during the checkout flow.

11 Global Hierarchy and Recognition

Presenting macro-level data like global rankings or national team profiles requires an architecture that is instantly recognizable and hierarchical.

11.1 Rankings and Visual Hierarchy

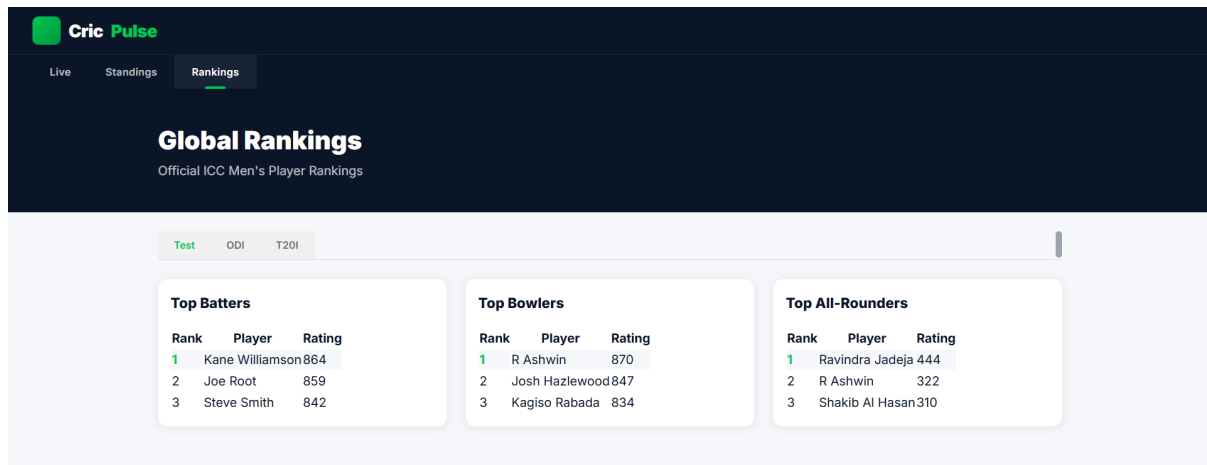


Figure 22: Rankings page establishing Visual Hierarchy

The ICC Global Rankings page must organize hundreds of players across multiple formats. The UI manages this through a strict Visual Hierarchy. The Number 1 ranked player in any category is decoupled from the standard table structure; they are highlighted with a distinct background color, larger font size, and an accent-colored rank number. This visual dominance instantly communicates their apex status. Secondary players follow in a standard, predictable list, ensuring the user can digest the hierarchical data instantaneously.

11.2 Team Profiles and Aesthetic Integrity

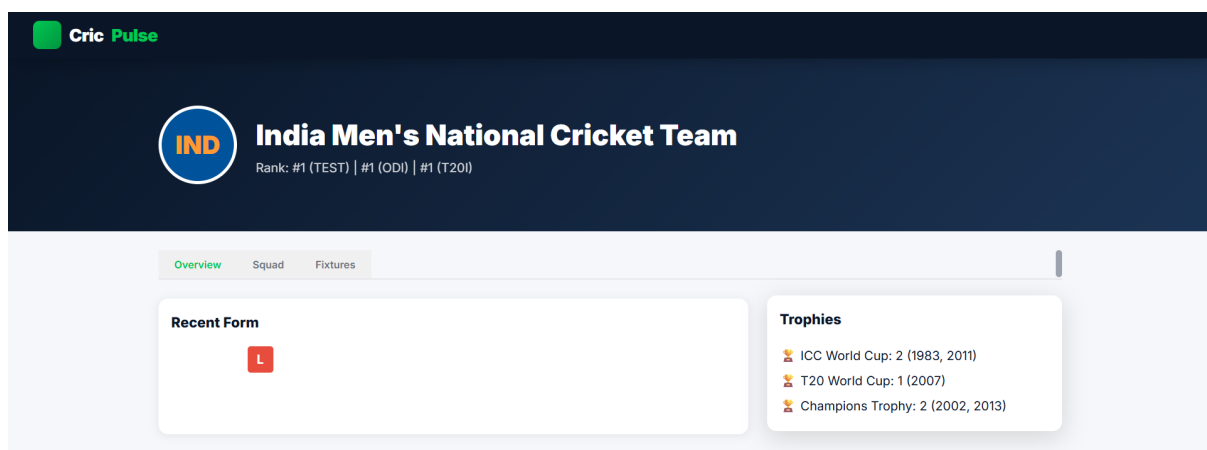


Figure 23: Team Profile page maintaining Aesthetic Integrity

The Team Profile page aggregates all data regarding a specific national team. The hero section utilizes Aesthetic Integrity by dynamically matching the background gradients and UI accents to the specific team's national colors (e.g., Blue and Orange for India). This validates the user's Mental Model and ensures they instantly recognize they are on

the correct page without needing to read the text. Below the fold, Tesler's Law is applied again, hiding the massive squad lists and fixture histories behind manageable, horizontal tabs.

Project Link

The fully interactive CricPulse portal can be accessed here: [View Live Prototype](#)